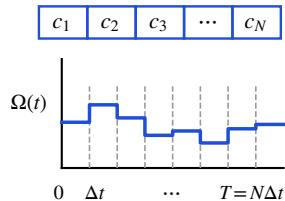
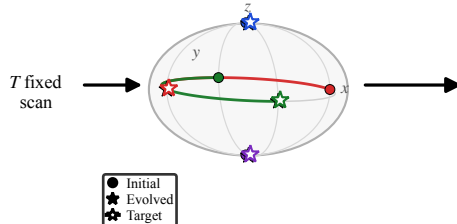


Traditional discrete optimization

(a) Pulse amplitudes $\{c_1, c_2, \dots, c_N\}$



(b) Propagator

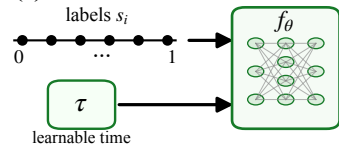


$$L_{\text{disc}} = \beta_{\text{gate}} L_{\text{gate}} + L_{\text{reg}}$$

terminal probes + pulse regularization

Continuous physics-informed representation

(c)



(d)

